# ORBITAL MEDICAL DIAGNOSTICS

MULTI-SYSTEM LABORATORY & PERFORMANCE REPORT

Patient: Alex Morgan | Record: SIM-261107-0042 | Date: 07/11/2026 | Time: 09:30 UTC

Mission day: FD-118 | Assessment window: 06–07/11/2026

## BONE HEALTH

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 1 | CTX / β-CTX | 1.82 | ng/mL |
| 2 | NTX | 115 | nmol BCE/mmol creatinine |
| 3 | Pyridinoline (PYD) | 165 | nmol/mmol creatinine |
| 4 | Deoxypyridinoline (DPD) | 38 | nmol/mmol creatinine |
| 5 | Helical Peptide | 220 | ng/mL |
| 6 | Hydroxyproline | 85 | mg/24 h |
| 7 | Hydroxylysine | 0.9 | mg/L |
| 8 | TRAP5b | 8.2 | U/L |
| 9 | P1NP / PINP | 12 | ng/mL |
| 10 | PICP | 22 | ng/mL |
| 11 | BSAP / BAP | 5.8 | µg/L |
| 12 | Osteocalcin | 3.5 | ng/mL |
| 13 | Serum Total Calcium | 10.1 | mg/dL |
| 14 | Urinary Calcium 24h | 480 | mg/24 h |
| 15 | Serum Phosphate | 4.8 | mg/dL |
| 16 | Urinary Phosphorus | 1.6 | g/24 h |
| 17 | PTH | 145 | pg/mL |
| 18 | 25(OH) Vitamin D | 12 | ng/mL |
| 19 | $1,25(OH)_2$ Vitamin D | 18 | pg/mL |
| 20 | FGF23 | 140 | RU/mL |
| 21 | Sclerostin | 58 | pmol/L |
| 22 | RANKL | 0.8 | ng/mL |
| 23 | OPG | 4.5 | pmol/L |
| 24 | DKK1 | 75 | ng/mL |
| 25 | Stable Ca isotope ratio | −0.8 | ‰ |
| 26 | Stable Ca tracer kinetics | 0.6 | fraction/day |
| 27 | Urinary Citrate | 110 | mg/24 h |
| 28 | Urinary Oxalate | 65 | mg/24 h |
| 29 | Urinary Magnesium | 55 | mg/24 h |
| 30 | Urinary Sodium | 220 | mmol/24 h |
| 31 | Urinary Potassium | 28 | mmol/24 h |
| 32 | Urinary Sulfate | 48 | mmol/24 h |
| 33 | Urinary Uric Acid | 910 | mg/24 h |

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 34 | Urinary Creatinine | 1,400 | mg/24 h |
| 35 | Urine pH | 5.1 | pH |
| 36 | Calcium Oxalate Supersaturation | 12.4 | index |
| 37 | Brushite / Ca Phosphate Supersaturation | 3.8 | index |
| 38 | Uric Acid Supersaturation | 2.9 | index |
| 39 | Sodium Urate Supersaturation | 3.5 | index |
| 40 | Struvite Supersaturation | 1.2 | index |
| 41 | Total Hip aBMD | 0.61 | g/cm² |
| 42 | Femoral Neck aBMD | 0.54 | g/cm² |
| 43 | Total vBMD | 95 | mg/cm³ |
| 44 | Cortical vBMD | 760 | mg/cm³ |
| 45 | Trabecular vBMD | 55 | mg/cm³ |
| 46 | Trabecular Number | 0.7 | 1/mm |
| 47 | Trabecular Separation | 2.1 | mm |
| 48 | Bone Failure Load | 2,100 | N |
| 49 | Estimated Bone Strength | 2,600 | N |
| 50 | Resistance Exercise Adherence | 2 | sessions/week |
| 51 | Resistance Exercise Load | 40 | % prescribed load |
| 52 | Dietary Calcium Intake | 420 | mg/day |
| 53 | Mission Duration | 118 | days |
| 54 | Microgravity Exposure | 118 | days |

## CARDIOVASCULAR HEALTH

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 55 | HR | 132 | beats/min |
| 56 | Resting HR | 132 | beats/min |
| 57 | Exercise HR | 156 | beats/min |
| 58 | HR recovery | 7 | beats/min decrease at 1 min |
| 59 | SBP | 88 | mmHg |
| 60 | DBP | 54 | mmHg |
| 61 | MAP | 65.3 | mmHg |
| 62 | Pulse pressure | 34 | mmHg |
| 63 | Stroke volume | 42 | mL/beat |
| 64 | Cardiac output | 5.54 | L/min |
| 65 | Cardiac index | 2.9 | L/min/m² |
| 66 | SVR | 1,100 | dyn·s/cm⁵ |
| 67 | CVP | 3 | mmHg |
| 68 | Plasma volume | 2.5 | L |
| 69 | Blood volume | 4.2 | L |
| 70 | Venous return | 4.8 | L/min |

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 71 | LV EDV | 86 | mL |
| 72 | LV ESV | 44 | mL |
| 73 | LV mass | 125 | g |
| 74 | LVEF | 49 | % |
| 75 | E/A | 0.6 | ratio |
| 76 | E/e′ | 19 | ratio |
| 77 | ECG | Sinus tachycardia | ECG finding |
| 78 | RR interval | 455 | ms |
| 79 | SDNN | 18 | ms |
| 80 | RMSSD | 12 | ms |
| 81 | HF HRV | 55 | ms² |
| 82 | LF HRV | 38 | ms² |
| 83 | PR | 120 | ms |
| 84 | QRS | 88 | ms |
| 85 | QTc | 510 | ms |
| 86 | PAC burden | 6 | % |
| 87 | PVC burden | 9 | % |
| 88 | AF burden | 2 | % |
| 89 | SVT/VT episode | 1 | episode/24 h |
| 90 | ST changes | −1.2 | mm |
| 91 | Orthostatic ΔSBP | −28 | mmHg |
| 92 | Orthostatic ΔHR | 42 | beats/min increase |
| 93 | Tilt tolerance | 4 | min |
| 94 | Baroreflex sensitivity | 3 | ms/mmHg |
| 95 | Cerebral flow velocity | 35 | cm/s |
| 96 | Orthostatic symptoms | 7 | score/10 |
| 97 | cIMT | 0.95 | mm |
| 98 | PWV | 12.4 | m/s |
| 99 | FMD | 2.5 | % |
| 100 | Carotid distensibility | 1.2 | %/10 mmHg |
| 101 | CAC score | 310 | Agatston units |
| 102 | Coronary CTA findings | 40 | % stenosis |
| 103 | Arterial plaque burden | 35 | % plaque volume |
| 104 | IJV peak velocity | 9 | cm/s |
| 105 | IJV flow grade | 3 | grade |
| 106 | IJV compressibility | 25 | % compression |
| 107 | Spontaneous echo contrast | Present | yes/no |
| 108 | Thrombus on ultrasound | Present | yes/no |
| 109 | D-dimer | 2.8 | mg/L FEU |
| 110 | Platelet count | 95 | ×10³/µL |

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 111 | Fibrinogen | 520 | mg/dL |
| 112 | PT/INR | 1.6 | ratio |
| 113 | aPTT | 42 | s |
| 114 | vWF | 195 | % activity |
| 115 | $VO_2$peak | 19 | mL/kg/min |
| 116 | Workload | 45 | W |
| 117 | $SpO_2$ | 91 | % |
| 118 | $PaO_2$ | 65 | mmHg |
| 119 | Respiratory rate | 26 | breaths/min |
| 120 | $EtCO_2$ | 28 | mmHg |
| 121 | Lactate | 4.2 | mmol/L |
| 122 | Sodium | 128 | mmol/L |
| 123 | Potassium | 5.8 | mmol/L |
| 124 | Magnesium | 1.3 | mg/dL |
| 125 | Calcium serum | 10.1 | mg/dL |
| 126 | Phosphate | 4.8 | mg/dL |
| 127 | Hemoglobin | 10.2 | g/dL |
| 128 | Hematocrit | 31 | % |
| 129 | Renin | 18 | ng/mL/h |
| 130 | Aldosterone | 38 | ng/dL |
| 131 | Vasopressin/copeptin | 45 | pmol/L |
| 132 | ANP | 35 | pg/mL |
| 133 | BNP/NT-proBNP | 820 | pg/mL NT-proBNP |
| 134 | Troponin I/T | 0.16 | ng/mL hs-cTnI |
| 135 | Glucose | 175 | mg/dL fasting |
| 136 | Insulin | 24 | µIU/mL |
| 137 | HbA1c | 9.1 | % |
| 138 | LDL-C | 188 | mg/dL |
| 139 | HDL-C | 28 | mg/dL |
| 140 | Triglycerides | 330 | mg/dL |
| 141 | ApoB | 160 | mg/dL |
| 142 | Lp(a) | 125 | nmol/L |
| 143 | hsCRP | 28 | mg/L |
| 144 | IL-6 | 74 | pg/mL |
| 145 | TNF-α | 42 | pg/mL |
| 146 | 8-OHdG | 35 | ng/mL |
| 147 | F2-isoprostanes | 1,800 | pg/mL |
| 148 | Cortisol | 31 | µg/dL at 08:00 |

## IMMUNE MONITORING

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 149 | WBC | 2.1 | ×10³/µL |
| 150 | Neutrophils absolute | 0.65 | ×10³/µL |
| 151 | Lymphocytes absolute | 0.8 | ×10³/µL |
| 152 | Monocytes absolute | 0.45 | ×10³/µL |
| 153 | Eosinophils | 0.12 | ×10³/µL |
| 154 | Basophils | 0.08 | ×10³/µL |
| 155 | Neutrophil-to-lymphocyte ratio | 0.81 | ratio |
| 156 | CD3+ T | 420 | cells/µL |
| 157 | CD4+ T | 95 | cells/µL |
| 158 | CD8+ T | 310 | cells/µL |
| 159 | CD4/CD8 | 0.31 | ratio |
| 160 | CD19+ B | 60 | cells/µL |
| 161 | CD16+/CD56+ NK | 45 | cells/µL |
| 162 | Naive/memory T subsets | 8 | % naive CD4 |
| 163 | Regulatory T cells | 2 | % of CD4 |
| 164 | T-cell activation CD69/CD25 | 6 | % CD69+ |
| 165 | T-cell proliferation | 18 | % divided |
| 166 | T-cell IFN-γ response | 12 | % IFN-γ+ |
| 167 | T-cell IL-2 response | 4 | % IL-2+ |
| 168 | NK cytotoxicity | 7 | % target lysis |
| 169 | NK perforin | 18 | % perforin+ |
| 170 | NK granzyme B | 12 | % granzyme B+ |
| 171 | NK CD107a degranulation | 5 | % CD107a+ |
| 172 | Monocyte cytokine response | 14 | pg/mL stimulated |
| 173 | Monocyte HLA-DR | 22 | % HLA-DR+ |
| 174 | Neutrophil oxidative burst | 15 | % DHR-positive |
| 175 | Phagocytosis index | 0.35 | relative index |
| 176 | Virus-specific T-cell frequency | 0.03 | % of CD8 |
| 177 | Virus-specific T-cell function | 4 | % cytokine-positive |
| 178 | Leukocyte bacterial challenge | 8 | % stimulated response |
| 179 | Antigen recall response | 3 | % stimulated response |
| 180 | IL-1β | 18 | pg/mL |
| 181 | IL-2 | 2 | pg/mL |
| 182 | IL-4 | 12 | pg/mL |
| 183 | IL-5 | 8 | pg/mL |
| 184 | IL-6 | 74 | pg/mL |
| 185 | IL-8/CXCL8 | 115 | pg/mL |
| 186 | IL-10 | 25 | pg/mL |
| 187 | IL-12 | 6 | pg/mL |
| 188 | IL-17A | 28 | pg/mL |

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 189 | IFN-γ | 35 | pg/mL |
| 190 | TNF-α | 42 | pg/mL |
| 191 | IFN-α | 18 | pg/mL |
| 192 | IFN-β | 15 | pg/mL |
| 193 | MCP-1/CCL2 | 310 | pg/mL |
| 194 | IP-10/CXCL10 | 850 | pg/mL |
| 195 | TGF-β | 19 | ng/mL |
| 196 | hsCRP | 28 | mg/L |
| 197 | Serum amyloid A | 95 | mg/L |
| 198 | Viral shedding duration | 12 | days |
| 199 | EBV-specific T response | 0.02 | % IFN-γ+ T cells |
| 200 | CMV-specific T response | 0.01 | % IFN-γ+ T cells |
| 201 | VZV-specific T response | 0.01 | % IFN-γ+ T cells |
| 202 | Viral symptoms | Present | yes/no |
| 203 | Cortisol serum | 31 | µg/dL at 08:00 |
| 204 | Cortisol saliva | 28 | nmol/L at 08:00 |
| 205 | Cortisol urine | 180 | µg/24 h |
| 206 | DHEA | 0.7 | µg/dL |
| 207 | Epinephrine | 210 | pg/mL |
| 208 | Norepinephrine | 780 | pg/mL |
| 209 | Sleep duration | 3.1 | h/night |
| 210 | Sleep efficiency | 52 | % |
| 211 | Circadian phase | 4.5 | h shift |
| 212 | Perceived stress | 31 | score/40 |
| 213 | Workload | 90 | NASA-TLX/100 |
| 214 | IgG | 380 | mg/dL |
| 215 | IgA | 32 | mg/dL |
| 216 | IgM | 25 | mg/dL |
| 217 | Salivary secretory IgA | 12 | mg/L |
| 218 | Vaccine antibody titer | 0.4 | IU/mL |
| 219 | 25(OH) vitamin D | 12 | ng/mL |
| 220 | Serum calcium | 10.1 | mg/dL |
| 221 | PTH | 145 | pg/mL |
| 222 | CTX | 1.82 | ng/mL |
| 223 | P1NP | 12 | ng/mL |
| 224 | Bone mineral density | 0.61 | g/cm² |
| 225 | Radiation dose | 180 | mSv cumulative |
| 226 | Microbiome diversity | 0.8 | Shannon index |
| 227 | Microbial pathogen detection | Detected | PCR result |
| 228 | Body temperature | 38.8 | °C |

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 229 | Symptom diary | Present | yes/no |

## BEHAVIORAL HEALTH

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 230 | Total sleep time | 3.1 | h/night |
| 231 | Time in bed | 6.0 | h |
| 232 | Sleep efficiency | 52 | % |
| 233 | Sleep onset | 02:10 | hh:mm |
| 234 | Wake time | 05:16 | hh:mm |
| 235 | Sleep onset latency | 92 | min |
| 236 | Wake after sleep onset (WASO) | 81 | min |
| 237 | Number of awakenings | 8 | count/night |
| 238 | Sleep midpoint | 03:43 | hh:mm |
| 239 | Sleep timing variability | 145 | min SD |
| 240 | Daily activity counts | 2,400 | counts/hour |
| 241 | Rest-activity regularity | 0.35 | index 0–1 |
| 242 | Subjective sleep quality | 1 | score 1–5 |
| 243 | Subjective sleep duration | 3.1 | h self-report |
| 244 | Nap duration | 0 | min |
| 245 | Sleep debt 7-day | 24 | h/7 days |
| 246 | PVT median reaction time | 610 | ms |
| 247 | PVT mean reaction time | 740 | ms |
| 248 | PVT lapses | 19 | count/10 min |
| 249 | PVT false starts | 12 | count/10 min |
| 250 | PVT fastest 10% RT | 410 | ms |
| 251 | PVT slowest 10% RT | 1,280 | ms |
| 252 | PVT reciprocal RT | 1.35 | $s^{-1}$ |
| 253 | PVT RT variability | 320 | ms SD |
| 254 | Subjective alertness VAS | 12 | VAS 0–100 |
| 255 | Subjective fatigue VAS | 88 | VAS 0–100 |
| 256 | Karolinska Sleepiness Scale | 9 | score 1–9 |
| 257 | Epworth Sleepiness Scale | 18 | score 0–24 |
| 258 | Motor Praxis speed | 695 | ms |
| 259 | Visual Object Learning accuracy | 17 | % correct |
| 260 | Fractal 2-Back accuracy | 17 | % correct |
| 261 | Abstract Matching accuracy | 8 | % correct |
| 262 | Line Orientation error | 24 | degrees error |
| 263 | Emotion Recognition accuracy | 12 | % correct |
| 264 | Matrix Reasoning accuracy | 12 | % correct |
| 265 | Digit Symbol Substitution correct | 22 | correct/min |

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 266 | Balloon Analog Risk Task choices | 38 | pumps/trial |
| 267 | Psychomotor Vigilance Task performance | 19 | lapses/10 min |
| 268 | Cognition task completion time | 780 | s |
| 269 | Cognition intraindividual change | −42 | % from baseline |
| 270 | Cognition practice-effect flag | Present | yes/no |
| 271 | Cognition missing-trial rate | 11 | % missing |
| 272 | Positive affect | 12 | score 0–100 |
| 273 | Negative affect | 86 | score 0–100 |
| 274 | Stress VAS | 88 | VAS 0–100 |
| 275 | Mental fatigue VAS | 91 | VAS 0–100 |
| 276 | Physical fatigue VAS | 87 | VAS 0–100 |
| 277 | Mood end-of-day | 12 | score 0–100 |
| 278 | Perceived coping | 15 | score 0–100 |
| 279 | Perceived recovery | 12 | score 0–100 |
| 280 | Motivation | 18 | score 0–100 |
| 281 | Loneliness | 85 | VAS 0–100 |
| 282 | Perceived social support | 10 | score 0–100 |
| 283 | Distress screening result | 18 | score 0–100 |
| 284 | Team cohesion | 12 | score 0–100 |
| 285 | Team conflict | 88 | score 0–100 |
| 286 | Team communication quality | 18 | score 0–100 |
| 287 | Team coordination | 12 | score 0–100 |
| 288 | Trust within crew | 18 | score 0–100 |
| 289 | Perceived privacy | 12 | score 0–100 |
| 290 | Habitability satisfaction | 15 | score 0–100 |
| 291 | Work-rest balance | 12 | score 0–100 |
| 292 | Interpersonal connectedness | 15 | score 0–100 |
| 293 | Ground communication satisfaction | 18 | score 0–100 |
| 294 | Team questionnaire completion | 100 | % complete |
| 295 | NASA-TLX mental demand | 96 | score 0–100 |
| 296 | NASA-TLX physical demand | 82 | score 0–100 |
| 297 | NASA-TLX temporal demand | 94 | score 0–100 |
| 298 | NASA-TLX performance | 12 | score 0–100 |
| 299 | NASA-TLX effort | 88 | score 0–100 |
| 300 | NASA-TLX frustration | 97 | score 0–100 |
| 301 | NASA-TLX overall | 92 | score 0–100 |
| 302 | Task error count | 18 | errors/day |
| 303 | Task completion time | 185 | % of baseline time |
| 304 | Task success rate | 42 | % successful |
| 305 | Operational response latency | 683 | ms |

| No. | Measurement | Result | Unit / scale |
|---|---|---|---|
| 306 | Schedule adherence | 55 | % adherence |
| 307 | Light exposure photopic | 35 | lux |
| 308 | Circadian-effective light | 15 | melanopic EDI lux |
| 309 | Noise exposure | 82 | dBA |
| 310 | Cabin CO2 | 5,200 | ppm |
| 311 | Cabin temperature | 29 | °C |
| 312 | Heart rate resting | 132 | beats/min |
| 313 | HRV RMSSD | 12 | ms |
| 314 | Salivary cortisol | 28 | nmol/L at 08:00 |
| 315 | Melatonin phase marker | 04:30 | hh:mm DLMO |
| 316 | Mission workload hours | 14 | h/day |
| 317 | Consecutive wake duration | 22 | h |
| 318 | Time since launch | 118 | days |
| 319 | Time to landing | 42 | days |
| 320 | Medication exposure flag | Present | yes/no |
| 321 | IPIP-NEO neuroticism | 92 | T-score 0–100 |
| 322 | IPIP-NEO extraversion | 18 | T-score 0–100 |
| 323 | IPIP-NEO openness | 20 | T-score 0–100 |
| 324 | IPIP-NEO agreeableness | 25 | T-score 0–100 |
| 325 | IPIP-NEO conscientiousness | 15 | T-score 0–100 |
| 326 | Baseline reliability | 0.45 | ICC |
| 327 | Within-person trend slope | −2.8 | z-score/month |
| 328 | Data completeness | 76 | % complete |